INCH POUND

MIL-PRF-3098/76F
11 July, 2011
SUPERSEDING
MIL-PRF-3098/76E
12 January 2005

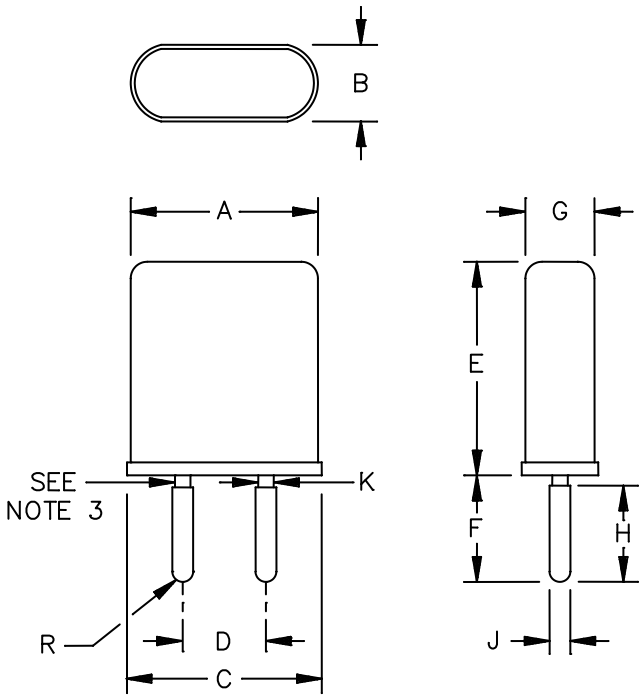
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR1/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 3.6 MHz to 10 MHz; fundamental; noncontrolled; antiresonance.



Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	1.125	---	28.58
B	---	.460	---	11.68
C	---	1.170	---	29.72
D	.490	.510	12.45	12.95
E	---	1.282	---	32.56
F	.625	.655	15.88	16.64
G	---	.411	---	10.44
H	.568	.588	14.43	14.94
J	.123	.127	3.12	3.23
K	.091	.095	2.31	2.41

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR1/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 3.6 MHz to 10.0 MHz, inclusive.

Mode of oscillation: Fundamental.

Rated drive level: 1.0 mW, maximum.

Operating temperature range (noncontrolled): -55°C to +90°C, inclusive.

Frequency tolerance: ±50 parts per million (ppm).

Equivalent resistance: See table I.

Antiresonance, load capacitance: 35.0 pF ±0.5 pF.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

TABLE I. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
3.6 to 5.5	90
5.5+ to 6.5	60
6.5+ to 7.8	50
7.8+ to 10.0	30

Part or Identifying Number (PIN): For Product Level B, PIN shall be CR25/U or CR25B/U (followed by specified frequency). For Product Level S, PIN shall be CR25S/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

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Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-PRF-202

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Custodians:
Army - CR
Navy - EC
Air Force - 99
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, CG, MC
Air Force - 19, 84

Preparing activity:
Army - CR

Agent:
DLA - CC

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NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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